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Neurodegenerative diseases, including Alzheimer's disease (AD), Parkinson's disease (PD), amyotrophic lateral sclerosis (ALS), frontotemporal dementia (FTD) and other related conditions, affect more than 57 million people worldwide[1]. Until recently, treatment options were limited to managing symptoms, but approvals for disease-modifying drugs for AD and genetic forms of ALS point to considerable progress[2,3]. Eventually, it may be possible to provide patients with targeted treatments, possibly in combination, that can prevent, slow, stop or reverse the progression of their disease[4–6]. However, several major obstacles have delayed the realization of this vision. First, many neurodegenerative conditions have an extended preclinical or prodromal period where diagnosis using available symptom-based assessments is either not possible or extremely difficult due to subtle manifestations that are not detectable by current clinical tools. Second, heterogeneity in the concordance between molecular pathology and clinical syndrome as well as common co-occurrence of multiple pathologies ('co-pathology') contribute to misdiagnosis in clinical settings. Third, additional variability exists in the rate and pattern of symptom progression within conditions, impeding efforts to accurately prognosticate disease course. These diagnostic and prognostic challenges ultimately hinder the efficacy of clinical trials and make successful treatment of patients with any approved disease-modifying therapy challenging.

Biomarkers have the potential to resolve some of these obstacles by enabling earlier diagnosis linked to pathological processes, providing methods to subtype diseases, predicting outcomes and ultimately guiding effective intervention[7,8]. They may also improve clinical trial design through precision recruitment and serve as pharmacodynamic or surrogate endpoints in experimental medicine. Recognizing this potential, the field has seen rapid advances in imaging and fluid biomarker research, leading to their growing incorporation into clinical trials and regulatory frameworks. Fluid biomarkers, in particular, offer a real-time window into brain pathology and may help bridge the long-standing disconnect between neuropathology and clinical symptoms in living patients. Reflecting this progress, clinical guidelines have begun to integrate fluid biomarkers into routine diagnostic workflows[9]. To date, fluid biomarker development has been most successful for AD, where markers of amyloid and tau pathology are now widely used. However, there remains an urgent need for reliable biomarkers of other neurodegenerative pathologies, including α-synuclein, TDP-43 and non-AD tauopathies. In addition, biomarkers that reflect non-specific but disease-relevant biological processes, such as neuroinflammation, metabolic dysregulation and vascular dysfunction, are essential to fully characterize the pathophysiological and molecular landscape of neurodegeneration.

The accelerating development of high-throughput molecular profiling technologies, combined with increasingly powerful computational tools applied to large, deeply phenotyped cohorts, is transforming the landscape of biomarker discovery[10]. Although multi-omics approaches, such as integration of genomics, transcriptomics and metabolomics, contribute to rich data-driven biomarker discovery, proteomics is uniquely positioned to impact both diagnosis and treatment of neurodegenerative disease. This is due to several key factors: (1) many clinically established biomarkers are protein based; (2) high-dimensional proteomic platforms such as SomaScan, Olink and mass spectrometry now offer sufficient depth to capture a sizable portion of the circulating proteome; and (3) protein-level changes often capture biological processes proximal to neurodegeneration, providing functional insights that are directly relevant to disease pathogenesis. Proteomic profiles derived from peripheral biofluids such as plasma and cerebrospinal fluid (CSF) not only hold promise for identifying biomarkers of disease presence and progression but also offer new avenues for therapeutic target discovery.

Robust high-dimensionality omics research in heterogeneous clinical groups necessitates the use of large datasets due to poor reproducibility of findings from single-site or smaller cohorts[11], but the siloing of data among a fragmented research community has been a barrier to such biomarker discovery[12]. Although many research institutions and initiatives have embarked on a variety of open data efforts, there is no standard model for providing researchers with easy access to data from multiple cohorts. Moreover, the use of such multicohort sources requires data aggregation and harmonization. The genetics research community has enabled huge consortia with joint data access and collaborative analysis[13]. In the field of neurodegenerative disease, large data-sharing efforts such as the ADNI[14], AMP-AD[15], PPMI[16], ALS TDI's ARC[17] and Answer ALS[18] are examples of open datasets that facilitate cross-study collaborations. However, despite these highly productive examples of best practice, their disease-specific design limits identification of shared mechanisms of neurodegeneration and potential co-existing pathologies. In addition, most data have either never been shared or have been kept behind restrictive barriers to access[12]. Reasons for this include a shortage of technology solutions, a range of challenging data governance rules and privacy regimes and cultural norms and misaligned incentives among researchers, research institutions, industry and research sponsors.

The GNPC was created to systematically address these challenges to large data analysis, accelerate biomarker discovery and advance the research and development of more precise treatments for neurodegenerative disease. Our goal was to generate a large proteomics resource using available samples from established cohort studies, accompanied by a harmonized clinical dataset, and to make these data available to the scientific community in a rapid and easily accessible manner. Having reached the planned point of public data release in July 2025, the GNPC has established what it thinks is the world's largest neurodegenerative disease-focused proteomics dataset for biomarker research, with 23 partners contributing more than 35,000 analyzed biosamples and approximately 250 million unique protein measurements with matched and harmonized clinical data. Here we summarize the GNPC version 1 (V1) dataset, together with key analysis vignettes. With the associated in-depth papers in this issue[19–21], this serves as the beginning of the explorations into this dataset and its contribution to the field of neurodegenerative disease.

Traditional 'on premises' data science analyses have become more challenging as datasets have increased in size, with a corresponding increase in resources required for moving data from one location to another. Moreover, local analysis presents a challenge in ensuring data integrity, safety and confidentiality. The GNPC's partnership with the AD Data Initiative provided the consortium with virtual access to the cloud-resident harmonized dataset with analysis workspaces via the AD Workbench[22], a secure, cloud-based environment that is able to satisfy multiple different geographical data jurisdictions (for example, the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA))[23].

For GNPC V1, we opted to use SOMAmer technology (provided by SomaLogic) as the primary proteomics platform as it was one of the broadest discovery platforms available. However, we also analyzed a subset of samples with Olink and mass spectrometry methods to allow for cross-platform comparison. In general, the GNPC's approach is agnostic to platforms and is guided primarily by coverage, reproducibility and affordability. We also think that different platforms bring complementary information as they may measure different isoforms and/or posttranslational modifications of a given protein.

Developing the GNPC's large dataset required addressing several barriers to open data and data sharing. To bring together data from several countries required navigating legal regimes with different requirements, including the GDPR (Europe), the Data Protection Act (United Kingdom) and HIPAA (United States). The GNPC's legal team worked with institutions in each jurisdiction to address specific concerns and agree on a framework for data sharing that worked across the board, including collecting the data on servers physically located in Western Europe.

Table 1 | GNPC contributing cohort details (n=23)

Cohort name	Organization	Geography (country)	Disease area	Sample collection and study methodology
Amyloid-beta in CSF for PD (ABC-PD)	University of Tübingen	Germany	PD	Link 1, Link 2
ALLFTD	University of California, San Francisco	United States	FTD	Link 1, Link 2
ALS Therapy Development Institute	ALS Therapy Development Institute	United States	ALS	Link 1
Alzheimer and Families (ALFA)	Barcelonaβeta Brain Research Center	Spain	AD	Link 1, Link 2
Amsterdam Dementia Cohort (ADC)	Amsterdam University Medical Center	The Netherlands	FTD and AD	Link 1, Link 2
Answer-ALS	Answer ALS	United States	ALS	Link 1, Link 2
Baltimore Longitudinal Study of Aging (BLSA)	NIA-NIH	United States	Aging	Link 1, Link 2
Banner Health	Banner Health	United States	AD and PD	Link 1, Link 2, Link 3, Link 4
BioFINDER	Lund University	Sweden	AD and PD	Link 1, Link 2
Bio-Hermes	Global Alzheimer's Platform Foundation	United States	AD	Link 1
CHARIOT-PRO	Imperial College London	United Kingdom	AD	Link 1
Emory ADRC	Emory University	United States	AD	Link 1
European Medical Information Framework Multimodal Biomarker Discovery Study (EMIF-AD MBD)	Maastricht University	The Netherlands	AD	Link 1, Link 2, Link 3
Fundacio ACE	Fundacio ACE	Spain	Mixed ADRD	Link 1, Link 2
Indiana ADRC	Indiana University	United States	Mixed ADRD	Link 1
Kansas ADRC	University of Kansas	United States	AD	Link 1
Knight ADRC	Washington University in St. Louis	United States	Mixed ADRD	Link 1, Link 2
Mayo Clinic Study of Aging (MCSA)	Mayo Clinic	United States	AD	Link 1
Parkinson's Progression Markers Initiative (PPMI)	Michael J. Fox Foundation	United States	PD	Link 1, Link 2
Religious Orders Study/Memory and Aging Project (ROSMAP)	Rush University	United States	AD	Link 1, Link 2, Link 3
Stanford ADRC	Stanford University	United States	AD	Link 1, Link 2
Tracking PD	Oxford University	United Kingdom	PD	Link 1
Whitehall II	University College London	United Kingdom	Aging, Mixed ADRD	Link 1, Link 2

All contributing cohorts to the V1 harmonized dataset include the name of the study, the contributing site or organization, the country where the study was conducted, the main disease area or focus of the study and the published sample collection protocols and study methodology. ADRC, Alzheimer's Disease Research Center; ADRD, Alzheimer's disease and related dementias.

The first version of the harmonized data was made available to consortium members in June 2024. Analysis of the harmonized dataset is organized in four workstreams to allow members of the consortium to collaborate on areas of related interest: longitudinal profiling, cross-sectional profiling, proteogenomics and prediction modeling. Here we present the first set of analyses of the GNPC dataset, including the overarching summary analyses and, in accompanying papers, the work conducted in the GNPC workstreams during the first year of data availability.

Results

Initial findings from the GNPC harmonized dataset

The GNPC V1 harmonized dataset is focused on neurodegenerative diseases and, more specifically, on AD, PD, ALS, FTD and aging among 18,645 participants, drawn from 23 individual cohorts across a total of 31,083 unique peripheral plasma, serum and CSF samples, culminating in 35,056 unique proteomic assays and approximately 250 million individual protein measurements (Table 1 and Supplementary Table 1). Most of the proteomics characterization comes from the SOMAmer-based capture array (SomaScan version 4.1, version 4 and version 3 platforms), measuring approximately 7,000 (n = 26,458), 5,000 (n = 4,528) or 1,300 (n = 95) unique aptamers per biosample, respectively. Additionally, 1,975 of the plasma samples characterized on the version 4.1 SomaScan platform had tandem mass tag mass spectrometry performed. The harmonized dataset additionally includes 40 clinical features, including demographic data, vital signs data and clinical features collected with each blood or CSF draw (Supplementary Table 3). These aggregated and harmonized data demonstrated their value to the consortium immediately, as they served to rapidly confirm signals originally identified in smaller datasets across the entirety of the GNPC V1 dataset, thereby serving as an 'instant validation' resource[24,25].

To evaluate the structure and comparability of the blood-based proteomics data, we conducted a principal component analysis on plasma and serum samples (Supplementary Fig. 1). Serum samples clustered distinctly from plasma, reflecting a clear matrix effect. Among plasma samples, a modest offset was observed between the 5K and 7K SomaScan platforms, whereas EDTA and citrate plasma samples appeared largely similar. All three vignettes described below focused exclusively on plasma proteomic data, and, where relevant, platform-related differences between 5K and 7K assays were addressed using scaling or predictive modeling approaches.

To highlight the breadth and utility of the GNPC V1 resource, we present three illustrative vignettes that showcase how this harmonized dataset can be applied to address key questions in neurodegenerative disease and aging research: (1) disease-specific differential abundance profiling, (2) biological aging across organ systems and (3) protein markers of genetic risk as exemplified by the apolipoprotein E (*APOE)* genotype. As summarized below and described in more detail in the accompanying papers in this issue[19–21], these vignettes reflect the analytical depth enabled by the GNPC and are intended to catalyze further exploration by the broader research community upon public data release.

Vignette 1: Human blood proteomic profiles are robustly associated with neurodegenerative diseases and clinical severity. We examined the plasma proteome as measured using the SomaLogic 7K platform, among people with AD, PD, FTD and ALS (referred to hereafter as 'Patients', $n = 3,002$), and separately among people with no neurodegenerative disease diagnosis and cognitively normal test screenings (referred to hereafter as 'Controls', $n = 5,879$) (see Vignette 1 methods for sample selection criteria). First, we sought to identify proteins differentially abundant in the plasma of patients with different neurodegenerative diseases with cognitive effects—namely, AD, PD, FTD and ALS. Leveraging the breadth of cohorts included in the GNPC, we first performed cohort-stratified analyses to internally validate the most robust protein changes, focusing on those consistently altered across multiple study cohorts. Cohort-stratified results were subsequently combined using a meta-analysis for AD (Fig. 1a), PD (Fig. 1b), FTD (Fig. 1c) and ALS (Fig. 1d).

In AD ($n = 1,966$), 27 proteins from AD Patients robustly emerged as being significantly elevated compared to Controls across at least six of the 10 different cohorts, including ACHE, SPC25, LRRN1 and CTF1. Additionally, GDF2 and APOB also showed high meta-analytic effect sizes and were independently significant in five and four separate cohorts, respectively (Fig. 1a). In contrast, 130 proteins were consistently lower in AD plasma across at least six cohorts, including VAT1, GPD1, ARPC2 and PA2G4. Furthermore, we observed significant decreases in RPS12, NPTXR and NT5C across five cohorts. These top hits highlight both expected and underexplored targets consistently altered in AD plasma across cohorts, including those with established ties to lipid metabolism (APOB and GPD1), cholinergic signaling and/or treatment response (ACHE and VAT1) and synaptic integrity (NPTXR), as well as novel targets linked to cytoskeletal regulation (ARPC2) and RNA metabolism (PA2G4 and RPS12). Follow-up analysis (see Vignette 3) also indicated that elevation of some targets, such as SPC25, LRRN1 and CTF1, reflected underlying *APOE* ε4 genotype effects rather than AD diagnosis per se. Reactome pathway analyses ($n = 2,640$, Bonferroni-adjusted $P < 0.05$) revealed enrichment for terms related to sugar metabolism ('glucose metabolism' and 'glycolysis') and protein prenylation ('RAB geranylgeranyltransferase'), reinforcing links to bioenergetics and vesicle trafficking (Fig. 1e).

In PD ($n = 607$), 40 proteins were significantly elevated across at least three of seven different cohorts, including SUMF1, PRR15, AARDC3 and RDH16, which were elevated in at least four cohorts (Fig. 1b). Additional proteins such as PSMC5, DDX1 and VSIR exhibited strong effect sizes and replicated in two cohorts. In contrast, 15 proteins were significantly lower in PD plasma across at least three cohorts, including CLEC3B, GPD1 and SEMA4G. Meta-analytic effect sizes were very high for PRSS8, BAGE3, NPS, PRL and HEXB, all of which decreased in PD but were less reproducible across contributing cohorts, suggesting cohort-specific factors driving depleted abundance of these targets. These candidate PD-associated proteins include targets associated with proteostatic (SUMF1, HEXB and PSMC5), immune (VSIR and CLEC3B) and axonal guidance (SEMA4G) pathways, possibly reflecting both peripheral and brain-related pathophysiology. Similar to in AD, Reactome pathway analyses ($n = 2,251$, Bonferroni-adjusted $P < 0.05$) revealed enrichment for terms related to Ras superfamily/small GTPases and vesicle trafficking ('vesicle-mediated transport' and 'signaling by RHO GTPases'), highlighting a plasma proteome pathway overlap between AD and PD (Fig. 1f).

Although FTD clinical syndromes are less common than AD or PD and have greater clinical and neuropathological diversity, nine targets exhibited decreased abundance in FTD plasma ($n = 175$) across multiple cohorts after Bonferroni correction (Fig. 1c). Strongly downregulated hits included NPTXR, APLP1 and HS6ST3, which converge on processes critical for synaptic maintenance and neuronal support. Eleven proteins were significantly elevated in FTD plasma with a conventional false discovery rate (FDR) correction but did not survive Bonferroni correction. Despite limited power, Reactome pathway analyses ($n = 71$, FDR < 0.05) revealed two significantly enriched terms, 'posttranslational protein phosphorylation' and 'regulation of insulin-like growth factor transport and uptake by IGFBPs', highlighting conserved peripheral signatures of neurodegeneration even amid the clinical and pathological heterogeneity of FTD (Fig. 1g).

In ALS ($n = 254$), we analyzed plasma proteomic profiles from a single contributing cohort with Patients and Controls (Fig. 1d). After FDR correction, 44 targets exhibited significantly increased abundance in ALS, including a host of proteins related to skeletal muscle structure and function (PDLIM3, MYOM2, MYLPF and TNNI2). Thirty-eight targets exhibited significantly decreased abundance in ALS, including two aptamers targeting ART3, an ADP-ribosyltransferase enriched in skeletal muscle, as well as additional proteins linked to growth factor signaling and/or extracellular matrix composition (ANTXR2, CRTAC1 and RGMA). Reactome pathway analyses ($n = 82$, FDR < 0.05) confirmed this strong biological enrichment for skeletal muscle-related processes ('muscle contraction' and 'collagen chain trimerization') (Fig. 1h), underscoring a clear peripheral proteomic footprint of ALS consistent with its primary motor system pathology.

After identifying disease-related differential abundance patterns, we combined data across AD, PD and FTD to identify a global signature of dementia severity. Specifically, a 256-protein clinical impairment signature was derived using least absolute shrinkage and selection operator (LASSO)-based prediction of Clinical Dementia Rating (CDR) global scores, which was subsequently evaluated in a held-out test set (2,047 records; 30% of the dataset). Signature values correlated with CDR global scores (train: Pearson's $r = 0.68$; test: $r = 0.58$) in a stepwise fashion, increasing at each level of clinical severity (Fig. 1i; see Supplementary Table 7 for individual-level scores). To demonstrate concordance with an orthogonal clinical outcome, we observed that higher signature values were also associated with lower cognitive test scores (standardized Montreal Cognitive Assessment/Mini-Mental State Examination (MoCA/MMSE); $r = -0.47$), consistent with expected inverse correlations between cognition and CDR. In disease-stratified analysis combined across training and test sets, the multivariate protein signature was reliably elevated with greater clinical severity in AD ($r = 0.55$), FTD ($r = 0.85$) and PD ($r = 0.70$), supporting its relevance as a transdiagnostic marker of clinical severity (Fig. 1j). Top proteins with high feature importance in the clinical impairment signature (Fig. 1k and Supplementary Table 8) again highlighted ACHE and NPTXR as well as additional targets linked to neuroplasticity (EPHA4 and CNTFR) and immune activation (MSMP and KLK3), underscoring their potential for transdiagnostic dementia staging.

Vignette 2: Organ age analysis reveals neurodegenerative disease-specific patterns of premature aging. One advantage of plasma proteomics is the ability to simultaneously query the health of distinct organ systems. We applied previously validated plasma proteomic organ aging models[26] to assess accelerated organ-specific aging across multiple Patient and Control data in the GNPC. Predicted organ ages showed moderate to strong correlations with chronological age (r range, 0.36–0.92; Fig. 2a), confirming model performance in the GNPC cohort[26]. Figure 2b shows the association between organ age gaps, which capture person-specific differences between estimated organ age and actual age, and AD, FTD and PD, respectively. Elevated cognition-enriched brain age gaps, reflecting the subset

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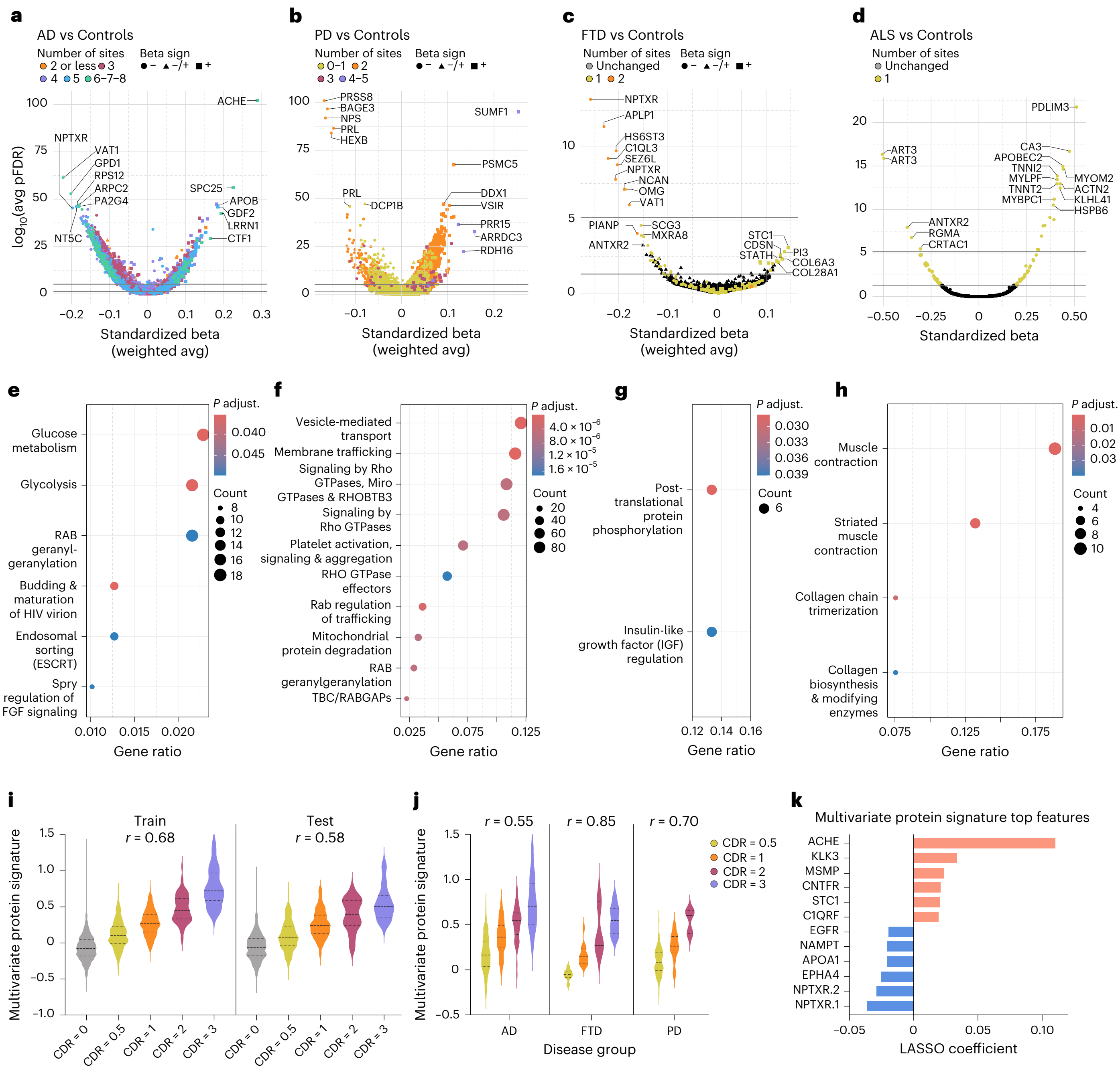

Fig. 1 | Circulating blood proteome specifies neurodegenerative disease type, mechanism and clinical severity. **a**–**d**, Meta-analytic differential abundance analysis showing changes in relative protein expression of AD (**a**), PD (**b**), FTD (**c**) and ALS (**d**) compared to Controls. Each dot represents a protein. The *x* axis shows the direction and effect size of protein changes relative to Controls, from linear regression models including age and sex as covariates; the *y* axis shows the $-\log_{10}$ FDR-adjusted *P* value. *P* values from two-sided tests and after adjustment from FDR are reported. The parallel line at the bottom of each plot shows which proteins are significant after FDR correction for multiple comparisons. The line above shows proteins further surviving Bonferroni correction. Dots are colored based on the number of cohorts where the protein was found to be independently significant after (within-cohort) FDR correction and changed in the same direction relative to Controls (that is, increased or decreased compared to Controls). **e**–**h**, Significant proteins from the differential abundance analyses were fed into Reactome enrichment analysis for AD (**e**), PD (**f**), FTD (**g**) and ALS (**h**), using unique SomaScan 7K proteins as background. Enriched Reactome pathway terms for each condition are visualized as dot plots, with dot size corresponding to the number of differentially abundant proteins assigned to a given pathway (one-sided Fisher's test with FDR adjustments). Full Reactome enrichment summary statistics are reported in Supplementary Table 6. HIV, human immunodeficiency virus; FGF, fibroblast growth factor; TBC/RABGAPs, Tre2–Bub2–Cdc16 (TBC) domain-containing RAB-specific GTPase-activating proteins. **i**, Violin plots displaying LASSO-derived clinical severity protein signatures across CDR global level in training and test sets. **j**, Violin plots displaying LASSO-derived clinical severity protein signatures across CDR global level (0.5 and higher) in AD, FTD and PD, using the combined training and test sets. **k**, LASSO coefficients for the top 12 protein aptamers selected in the clinical severity protein signature. avg, average; pFDR, FDR-corrected *P* value.

of brain-specific proteins that previously enhanced model age gap prediction of cognitive impairment, were associated with higher odds of AD (odds ratio = 1.33 per 1-s.d. age gap increase (95% confidence interval: 1.25–1.41)) and FTD (odds ratio = 1.26 (95% confidence interval: 1.06–1.48)). Non-cognition-enriched brain age gap was weakly associated with AD risk (odds ratio = 1.08 (95% confidence interval: 1.02–1.14))

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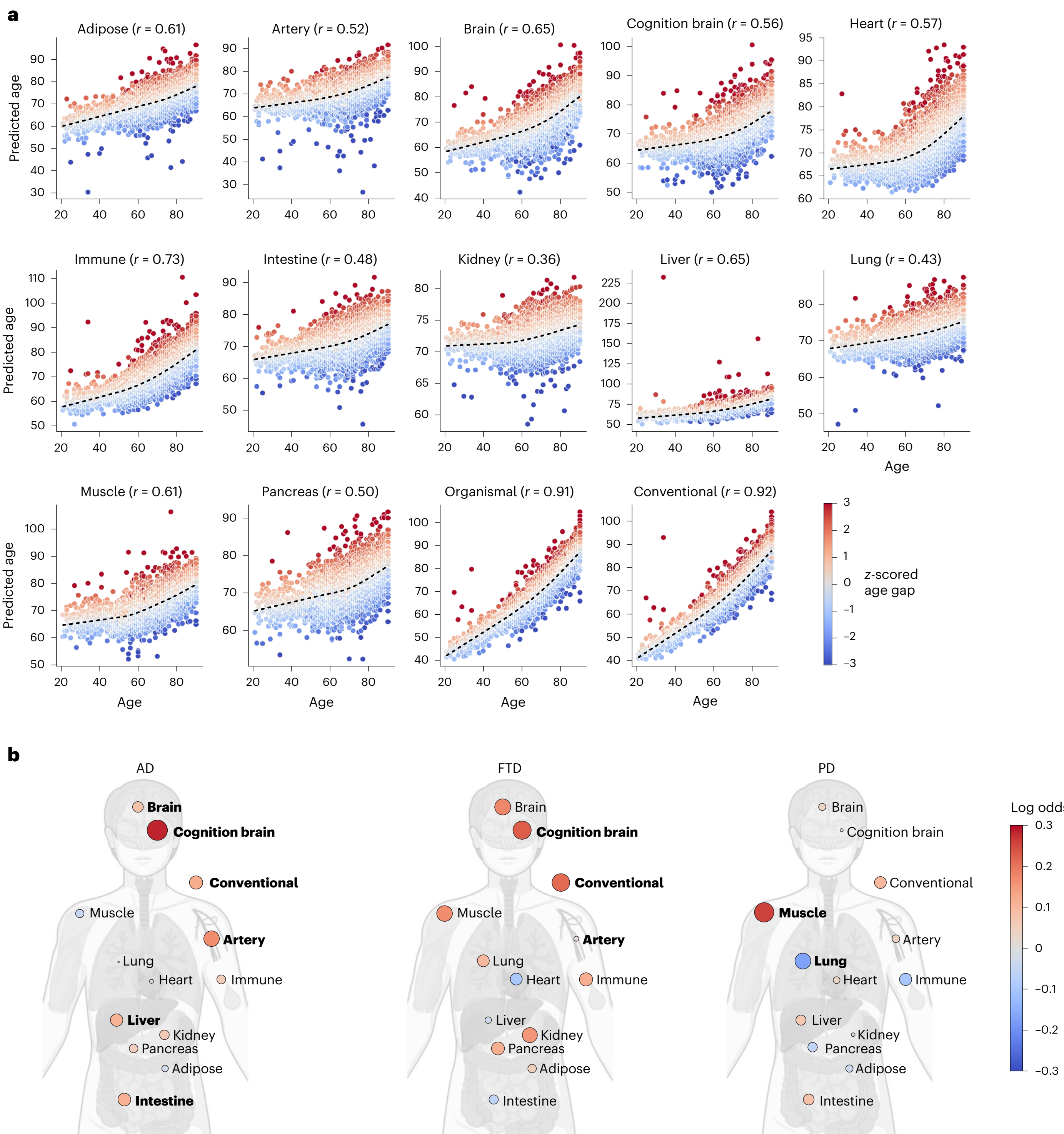

Fig. 2 | Organ age patterns characterize distinct neurodegenerative disease types. a, Scatterplots of chronological age versus predicted age for each organ aging clock in clinically normal individuals. Black dashed line indicates the LOWESS regression estimate of the population mean. Pearson's correlation coefficient *r* is reported for each clock. **b**, Body plots showing associations of standardized organ age gaps with neurodegenerative disease based on binary logistic regression models. *P* values are from two-sided tests. Red dots indicate positive associations (higher age gap with disease); blue dots indicate negative associations (lower age gap with disease). Bold labels highlight organ ages associated with organ age gap with $P < 0.05$ after FDR correction. The body plots were created in BioRender: Oh, H. (2025): https://BioRender.com/afoqtwz.

but not other conditions. Beyond brain, we observed contributions of artery (odds ratio = 1.18 (95% confidence interval: 1.11–1.25)), liver (odds ratio = 1.11 (95% confidence interval: 1.05–1.17)) and intestine aging to AD (odds ratio range, 1.12–1.18) as well as a unique link between muscle aging and PD (odds ratio = 1.12 (95% confidence interval: 1.05–1.19)). These findings extend previous work by demonstrating shared and distinct patterns of blood-detectable accelerated organ aging across AD, FTD and PD, underscoring connections between systemic health and neurodegenerative disease that may be related as a cause, correlate or consequence.

Vignette 3: Human blood proteomic signatures reflect *APOE* genetic status and uncouple systemic AD and *APOE* effects. Using a combination of machine learning and biological enrichment approaches, we sought to isolate the molecular signatures of *APOE* ε4, the main genetic risk factor for sporadic AD, independent of AD and other conditions. Several proteins, including SPC25, LRRN1, S100A13 and NEFL, were strongly associated with *APOE* ε4 versus other alleles (Fig. 3a), paralleling prior observations for some of these targets in serum[27]. Some proteins, such as SPC25 and LRRN1, showed no difference between AD and Controls but exhibited dose-dependent associations with *APOE* ε4 (Fig. 3b), suggesting that previously identified links to AD from Vignette 1's differential abundance analysis were driven by *APOE* ε4 enrichment in AD cases. Conversely, well-known AD-associated neuromodulatory proteins such as NPTXR and GDF2 were robustly associated with AD diagnosis, irrespective of *APOE* ε4 allelic dose (Fig. 3c).

Notably, the effects of *APOE* ε4 genotype on the plasma proteome were so robust that a machine learning model with only five proteins (SPC25, NEFL, S100A13, TBCA and LRRN1) predicted *APOE* ε4 carrier status in unseen patients with high accuracy, both within AD and within non-AD Patients (area under the curve (AUC) range, 0.90–0.96; Fig. 3d). Leveraging protein–protein interaction libraries and single brain cell RNA sequencing data from the Human Protein Atlas[28], we observed that three of these proteins (SPC25, TBCA and S100A13) were central nodes in the protein–protein interaction network (Fig. 3e) and brain cell type expression patterns (Fig. 3f)[28]. Ubiquitin-C, however, was found to be a key connection point across the central node proteins, suggesting potential convergence on proteostatic pathways.

Lastly, to identify potential genotype–phenotype links, we compared proteins associated with *APOE* ε4 in cognitively unimpaired individuals (n = 2,817; 215 proteins) to those associated with AD in *APOE* ε3 homozygotes (n = 1,843; 2,150 proteins). Forty-four overlapping proteins showed consistent directionality (Fig. 3g), including targets elevated in *APOE* ε4 and AD. Patients involved in immunovascular signaling (MMP8), synaptic vesicle fusion (SNAP23) and lipid trafficking (APOB) pathways relevant to *APOE* biology and cognitive decline. This overlap highlights potential early molecular footprints of AD pathophysiology present even in asymptomatic *APOE* ε4 carriers and underscores the utility of contextualizing by *APOE* genotype when seeking AD-relevant proteomic signals. These shared proteins may reflect core features of *APOE*-related biology that are also prominent in AD, highlighting potential mechanisms through which *APOE* ε4 contributes to disease vulnerability.

Discussion

The availability of high-dimensional molecular datasets has led to an increasing number of large-scale collaborative programs to share and use these data—a trend forged by the genetics community. Following these data-sharing collaborations, numerous initiatives have led the way in data sharing with the wider scientific community. We would like to highlight in particular two programs focused on neurodegeneration or proteomics. The first large-scale open data-sharing program in neurodegeneration was the Alzheimer's Disease Neuroimaging Initiative (ADNI): an immensely productive public–private partnership that continues today and has spawned many followers. These include AddNeuroMed/InnoMed, an ADNI-like program in Europe that served as a pilot for the European Union Innovative Medicines Initiative (IMI) funding scheme that itself has generated many data-sharing and sample-sharing programs, including, for example, IMI-EMIF and IMI-EPND in neurodegeneration. In the proteomics arena, the GNPC was preceded by the UK Biobank Pharma Proteomics Project (UKBB-PPP) that generated extensive protein data on 50,000 research participants and is now planning analysis on an additional 250,000 participants, to accompany the extensive clinical, imaging and genomic data available to the scientific community. The GNPC complements both initiatives and many others in providing a disease-focused dataset, as in the ADNI, but at scale, as in the UKBB-PPP.

Notably, a key point of emphasis of the GNPC's mission is the strong intent to share this dataset with the global research community early in its life cycle. We do not present these analyses as definitive but, rather, as the pilot experiments by a subset of researchers whose datasets contributed to the construction of GNPC V1. We hope this summary paper and more in-depth papers serve as an invitation of collaboration and/or independent analysis from the global community. Only then will we be able to maximize disease insights from GNPC V1 and its combination with other datasets to accelerate translation of insights into the next generation of diagnostics and therapeutics for neurodegenerative diseases.

The initial analyses presented here underscore the versatility and translational potential of the GNPC dataset. First, disease-specific differential abundance and disease-shared clinical severity analyses revealed both established and novel protein targets in plasma across AD, PD and FTD, highlighting shared and distinct biological processes such as vesicle trafficking, synaptic integrity and metabolic dysregulation. These results not only validate previously reported protein markers but also nominate new candidates for mechanistic follow-up and blood biomarker development—an urgent clinical need across diseases. Second, organ aging clocks applied to the GNPC dataset uncovered disease-specific patterns of accelerated aging across brain and peripheral organs, offering a systems-level view of proteomic aging that bridges central and systemic health. These findings extend previous work on biological aging by demonstrating that distinct conditions are associated with unique organ-specific age gaps, supporting their relevance to age-related neurodegenerative diseases. Finally, proteome-wide analysis of the *APOE* genotype revealed a robust and disease-independent *APOE* ε4 signature, with potential mechanistic relevance to proteostasis and lipid transport. These vignettes, each explored further in companion publications[19–21], illustrate the range of insights made possible by GNPC V1 and set the stage for future hypothesis-driven and exploratory research by the broader scientific community, with the potential to better support trial design, monitoring and subtyping of clinical patients.

These data, as well as these vignette analyses and those in the accompanying papers, suggest that very large protein datasets have potential to add value to drug discovery. Hitherto, the value of genetic data has been increasingly recognized in supporting effective drug discovery. Notably, targets with genetic support are more likely to progress through the drug discovery pipeline[29] with probability of success recently being calculated to be 2.6 times greater than in targets lacking genetic support, even for genes with small effect sizes[30]. In contrast to the inherited traits represented by genetic variants, proteomics represents biological states. Although genetic factors are intrinsically causal in their relationship with disease, a proteomic association with disease might be consequential of the disease, a factor associated with disease (including response to a therapeutic) or reflect a causal process. Although these observations might suggest a role for proteomics more in supporting drug discovery through biomarker discovery rather than for target identification, the vignettes reported here and in the accompanying papers illustrate the potential for more direct drug target identification/validation. For example, in Vignette 1, the proteomic profiles identified include strong support for synaptic dysfunction with proteins identified that are already clearly part of a mechanism targeted for neurodegeneration drug discovery, such as NPTXR[31], whereas, in Vignette 3, proteins are identified that are very strongly associated with *APOE* status and with disease state. These proteins will surely now be considered as possible targets for drug discovery, especially considering that the *APOE* genotype is a striking example of a very strong genetic risk factor that has not been the source of equally strong drug discovery programs. The GNPC data also support

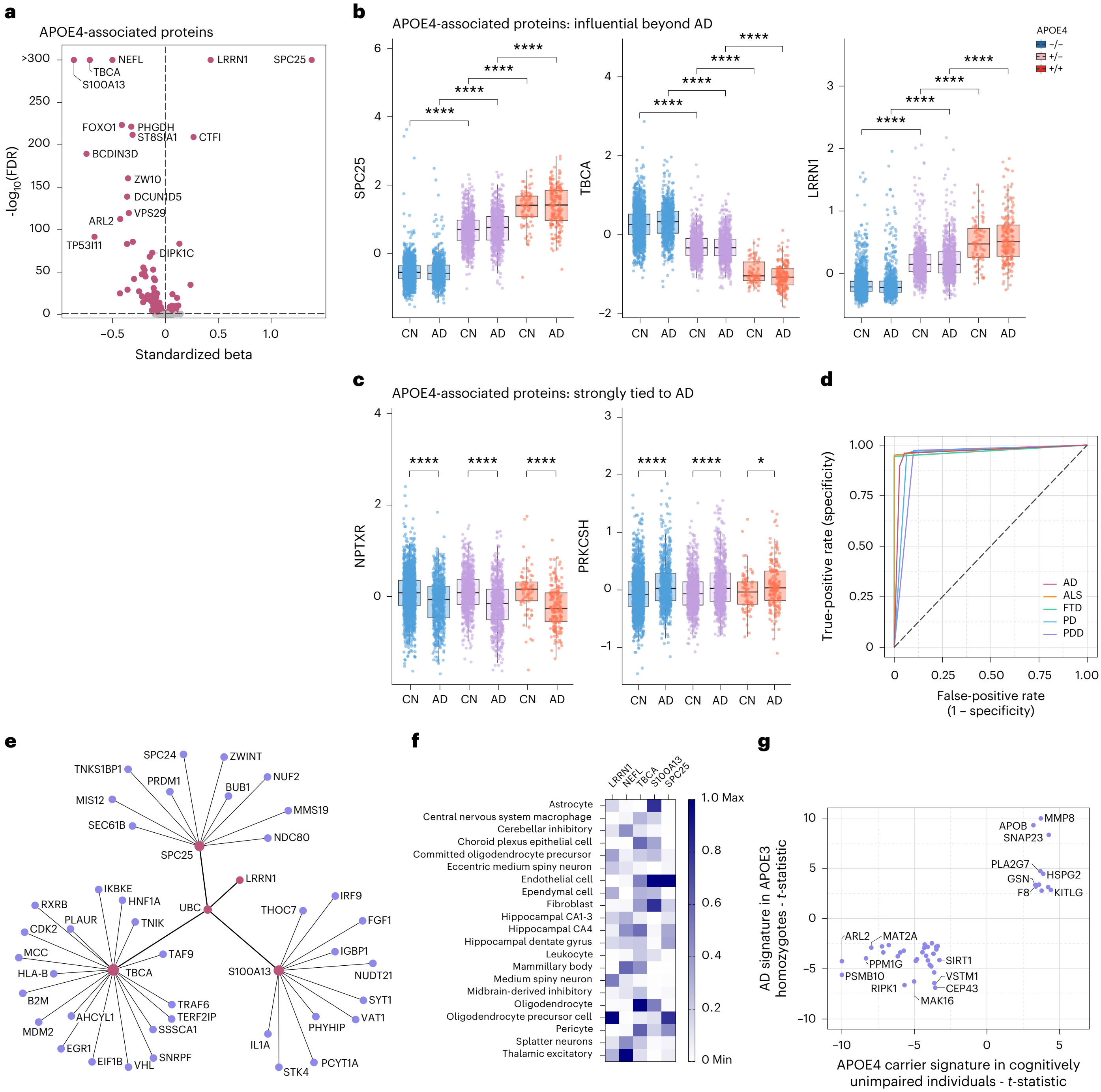

Fig. 3 | Disease-dependent and disease-independent of *APOE* ε4 on the human proteome. a, Volcano plot shows the protein association profile of *APOE* ε4 after adjusting for AD dementia diagnosis, with red representing significant associations (after FDR correction). At the *y* axis, the $-\log_{10}$(FDR-adjusted *P* values) > 300 were set to 300 for better visualization. This was done for S100A13, TBCA, NEFL, LRRN1 and SPC25. **b**,**c**, Box plots show plasma protein level changes of the proteins with the strongest *APOE* ε4 associations (**b**) and for *APOE* ε4-associated proteins strongly tied to AD dementia diagnosis (**c**). For **b** and **c**, the *y* axis represents residual protein levels after adjusting for age, sex, mean protein level and contribution site. The center line of each box indicates the median, with lower and upper edges representing the 25th and 75th percentiles. Whiskers extend to the most extreme values within 1.5 times the interquartile range; data points beyond this range were excluded as outliers. The *x* axis represents AD diagnosis. The color indicates *APOE* ε4 carrier status; '−/−' indicates *APOE* ε4 non-carriers; '±' indicates ε3/ε4; and '+/+' indicates ε4/ε4. Welch's *t*-test was used to compare residual protein levels between groups. Two-sided *P* values are reported. ****$P < 0.0001$ and *$P < 0.05$. *P* values were not adjusted for multiple comparisons, as only prespecified group contrasts are shown. Results marked with **** remain significant (pFDR < 0.0001) even after adjustment for multiple comparisons with the Benjamini–Hochberg method, whereas those marked with * do not. **d**, Receiver operating characteristic area under the curve (ROC-AUC) showing the performance of a machine learning model using only five proteins to predict *APOE* ε4 status across different diagnostic groups, in a held-out sample. **e**, Protein interaction network including four of those five proteins (red). **f**, Neural cell type expression of RNA transcripts encoding the five *APOE* ε4-predictive proteins. Plot shows mix-max scaling of protein-coding transcripts per million for each identified APOE ε4 protein. **g**, Correlation of effect sizes for proteins associated with *APOE* ε4 in cognitively unimpaired samples (*x* axis) and AD associated with AD diagnosis in *APOE* ε3/ε3 homozygotes. Limma *t*-statistic is shown for both contrasts; only proteins associated with both *APOE* ε4 and AD (adjusted $P < 0.05$ for both analyses) with the same direction of effect are visualized. For visibility purposes, *t*-statistic values higher than 10 were capped. PDD, Parkinson's disease dementia; pFDR, FDR-corrected *P* value.